



SQL Murder Mystery

Capstone Project

A data-driven investigation into a murder at TechNova Inc., using SQL to analyze keycard logs, call records, alibis, and evidence to uncover the truth.

The Crime Scene

Location

TechNova Inc. CEO's Office

Time

October 15, 2025

20:50 - 21:05

Challenge

Scattered logs, conflicting alibis, and hidden clues require systematic SQL analysis to solve.



Investigation Approach

01

Identify Crime Window

Determine who accessed the CEO's office during the murder timeframe.

02

Analyze Access Logs

Cross-reference keycard data with employee alibis to find inconsistencies.

03

Investigate Communications

Examine call records overlapping the critical time period.

04

Review Evidence

Catalog physical evidence found at the crime scene.

05

Connect the Dots

Identify suspects whose movements, alibis, and calls don't align.

Step 1: Who Was There?

First query identified employees who entered the CEO's Office during the murder window (20:50-21:05).

```
SELECT k.employee_id, e.name, k.room,  
       k.entry_time, k.exit_time  
FROM keycard_logs k  
JOIN employees e ON e.employee_id = k.employee_id  
WHERE k.room = 'CEO Office'  
AND k.entry_time <= '2025-10-15 21:05:00'  
AND k.exit_time >= '2025-10-15 20:50:00';
```

Result

David Kumar (Employee ID: 4) accessed the CEO Office from 20:50:00 to 21:00:00 — directly overlapping the murder timeframe.

Step 2: False Alibis Detected

Second query cross-referenced claimed locations against actual keycard logs to expose lies.

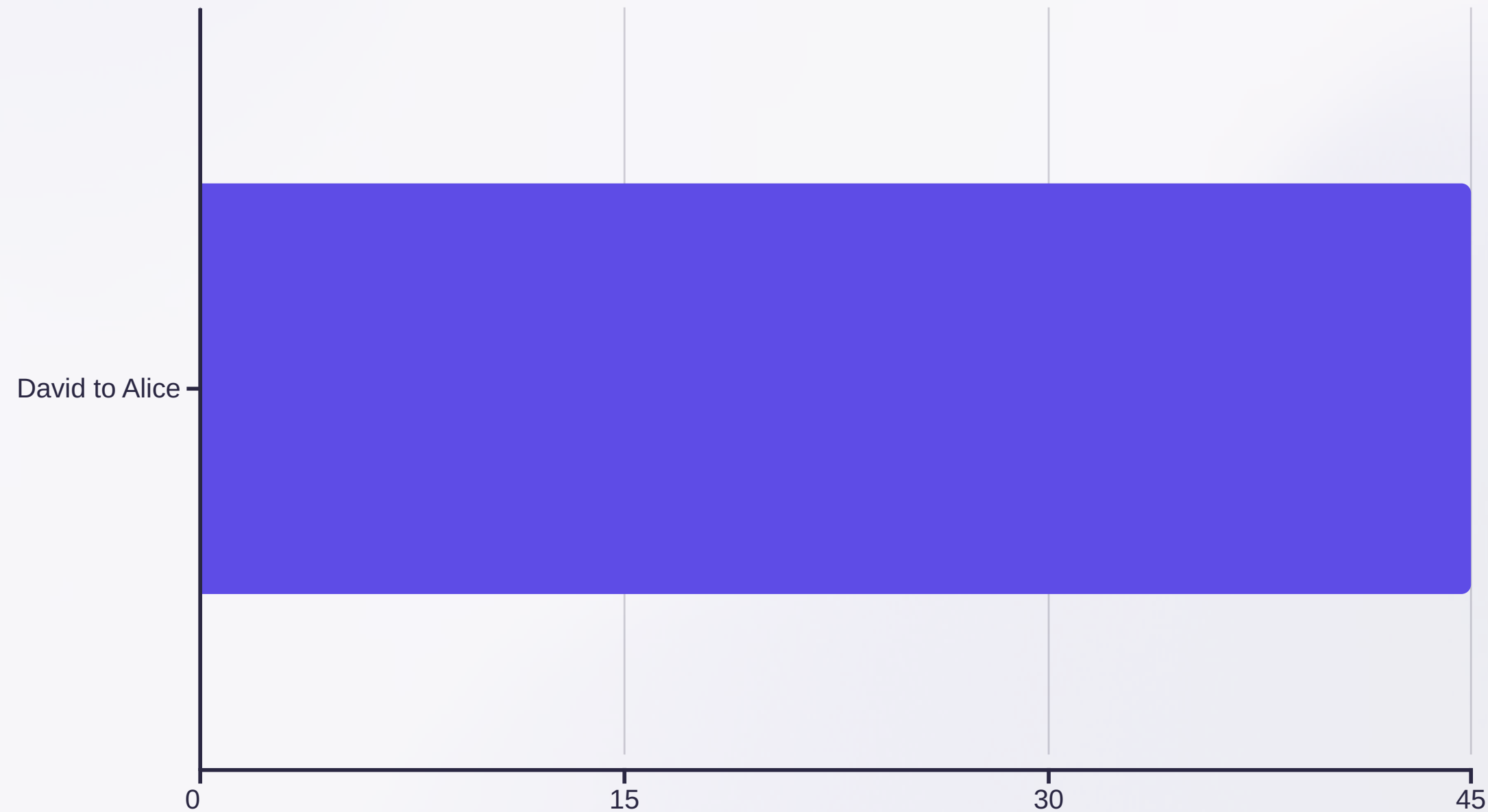
```
SELECT a.employee_id, emp.name,  
       a.claimed_location, k.room AS actual_room  
FROM alibis a  
JOIN keycard_logs k  
  ON a.employee_id = k.employee_id  
WHERE a.claimed_location <> k.room;
```

Alibi Mismatch

David Kumar claimed to be in the **Server Room** at 20:50, but keycard logs placed him in the **CEO Office**.

Step 3: Suspicious Call Activity

Third query revealed phone communications during the critical timeframe.



Call Record

David Kumar called Alice Johnson at 20:55:00 for 45 seconds — right in the middle of the murder window.

Step 4: Physical Evidence

Fingerprint on Desk

Found at 21:05:00 in CEO Office

Keycard Log Mismatch

Discovered at 21:10:00 in CEO Office

Evidence found at the crime scene pointed to access irregularities and physical presence at the desk.

Step 5: Connecting All Clues

Final query combined all evidence streams to identify the suspect whose story completely fell apart.

```
SELECT emp.employee_id, emp.name
FROM employees emp
WHERE emp.employee_id IN (
  -- Present at CEO Office during murder
  SELECT employee_id FROM keycard_logs
  WHERE room = 'CEO Office'
  AND entry_time <= '2025-10-15 21:05:00'
)
AND emp.employee_id IN (
  -- Provided false alibi
  SELECT a.employee_id FROM alibis a
  WHERE a.claimed_location <> k.room
)
AND emp.employee_id IN (
  -- Made calls during timeframe
  SELECT caller_id FROM calls
  WHERE call_time <= '2025-10-15 21:00:00'
);
```


Investigation Findings



Crime Scene Access

David Kumar accessed the CEO's office during the critical 20:50-21:00 window.



False Alibi

He claimed to be in the Server Room but was actually in the CEO Office.



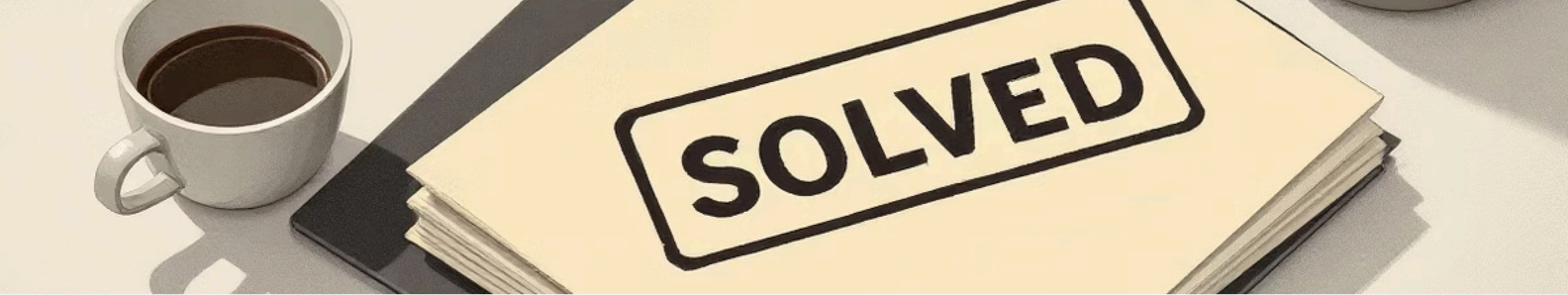
Suspicious Communication

He made a call to Alice Johnson at 20:55 during the murder timeframe.



Physical Evidence

Fingerprints and keycard mismatches place him at the scene with no valid explanation.



The Murderer

David Kumar

SQL analysis revealed David Kumar as the prime suspect. His presence at the crime scene, false alibi, suspicious call activity, and contradicting evidence logs all point to his guilt.

Case Status: SOLVED